

DATSET DESIGN AND QUALITY EXPECTATION

EXPECTED FILES AND STRUTURES

- Time series readings, Timestamped readings for each charging process: voltage, current, temperature, cumulative charge and SoC (when available).
- Session metadata/index One entry per session with details such as cell/pack identification number, chemistry, ambient temperature, charge process, start and end timestamps, and success flag.
- Cell/Pack metadata (if available), Cell level static information: capacity, chemistry, manufacturing date, number of cycles completed.
- Label/Annotation notes (if available): How the SoC ground truth is generated (from a reference cell, from an OCV chart from laboratory, or Coulomb-counting reference).

EXPECTED COLUMNS, LABELS & TARGETS

- Timestamp/ elapsed_time_s – Index
- Session_id – Identifier
- Cell_id – Identifier
- Voltage – feature
- Current – feature
- Cumulative_charge – feature

NAME: M A Niranjani Nair

- SoC_percent – target
- Cycle_number – metadata/feature
- Charge_rate – metadata/feature

Important metadata before using dataset:

- Units of each numeric column (voltages in volts/millivolts; time in seconds/milliseconds) – clearly indicated, rather than implied by the column name.
- Sampling rate/frequency/interval and if it is consistent across all sessions/data collection runs.
- Missing values representation used by the data provider (blank values, NaNs, or special sentinel values).
- Time zone of timestamp values and if the values are monotonic within a session.
- The reference technique used to generate SoC labels (reference cell, or coulomb counting integration) along with its accuracy.
- Any processing which was performed on the data by the data provider prior to file being obtained by this project.

Likely Risks – Missing data, Annotation, Imbalance, temporal, leakage

- ❖ Missing Data: Sensor failure during the session (missing temperature/current readings due to heavy load); Ground truth for SoC is available only for some sessions. Filling the gaps in silent fashion may lead to an artificial smooth curve being created, particularly at the very extremes of the charge cycle where the project assumes things would be difficult. Computing missing rate per column and per session; treating dropouts as gaps.
- ❖ Annotation Quality: The SoC label could also have its own drift, since it could be derived from Coulomb counting, rather than being a lab reference label. When a label is noisy, any model will not do better than that noise floor. The accuracy that is obtained should therefore not be

considered the ground truth but viewed relative to the error of the label itself.

- ❖ Imbalance: The majority of observations probably belong to the 30–80% SoC range, with a low density of observations in the ranges close to 0% and 100%. The very region where tree-based models struggle according to the literature is the one that is the least covered. Make sure to generate a histogram of the distribution of the SoC bands prior to model building; evaluate the performance per band, not in total.
- ❖ Temporal: Cells will deteriorate during the period of time in which the data is collected (capacity fade); the sampling rate could be different for each session or group of data collected. There could be calendar time effects learned by the model rather than charging effects if there was no separate tracking of cell aging.
- ❖ Leakage:
 - i. Split level- Row-wise splitting as opposed to session-wise splitting ensures that the very close correlated data from adjacent sessions ends up in both training and testing sets.
 - ii. Feature level- The features constructed using whole session data that are unavailable during prediction (total session duration, maximum capacity, or the accumulated charge column which is itself used to construct the SoC label).
 - iii. Preprocessing- These operations are performed before splitting the data into training and testing datasets.

Correct Train/ Validation/ Test Split

- Complete sessions, never divide them into rows. For a given session_id, all the rows associated with it must entirely belong to either the training, validation or test set, as confirmed by there being no overlap between the session_id across all three sets.
- The coverage with respect to SoC band, temperature, charge rate/protocol and chemistry (if mixed) should be evenly distributed among the splits, unless a deliberate temporal holdout is adopted — in

which case the discrepancy should be intentional and documented, not accidental.

- The choice to include a temporal holdout option was deliberate. If concerns about cell ageing exist, then it is advisable to train the model on sessions that were collected earlier and to test it on those that were collected later, in order to verify that the model generalizes to aged behavior and not just to sessions similar to those it was trained on.
- None of the preprocessing steps were applied outside the training set any scalers, imputers or feature selection steps were fitted only on the training portion and then used without change on both the validation and test sets.
- Reproducibility: the split proportions, the random seed (or the exact session-id lists for a temporal split), and the resulting session counts for each split are recorded so that the split can be regenerated in the same way later.
- The test set should include a sufficient number of sessions in the most difficult bands (those near 0% and 100% SoC) so that it is possible to give a meaningful per-band metric, rather than just an overall score which is dominated by the easier mid-range values.

Dataset Risk Checklist

- Schema & completeness: All files present (readings, session metadata, cell/label notes); columns match schema (units/types), mismatches resolved with data owner. Missing-value encoding documented; every session_id has a matching metadata row (no orphans)
- Missing data: Per-column missing-rate computed; pattern checked (random vs. concentrated by session/sensor/time). Sessions without SoC ground truth explicitly excluded/held out, not imputed
- Label quality: SoC ground-truth method (lab vs. Coulomb-count) and expected error documented per session/batch. Session outcome (completed vs. aborted) checked before deriving remaining-time labels. Sample of labels spot-checked against raw voltage/current curves
- Imbalance: SoC-band histogram computed, extremes (0–10%, 90–100%) checked. Temperature/charge-rate/chemistry distribution checked

across sessions. Imbalance plan agreed (per-band metrics/sampling, no naive oversampling)

- Temporal integrity: Sampling interval consistency checked per session; resampling plan defined if needed. Cycle_number/collection date available for aging-drift detection. Timestamps validated (timezone, strictly increasing per session)
- F. Leakage: No features derived from whole-session info unavailable at prediction time. Cumulative-charge/Coulomb-count columns checked against label derivation for direct leakage. Preprocessing fit on train split only; splits done at session level (empty session-id intersection check)
- G. Split validation: Train/validate/test counts/proportions documented. SoC-band/temp/chemistry distribution compared across splits, differences explained. Split reproducible (fixed seed or recorded session-id list)